

Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in $M \geq 6.5$ Earthquake Catalogs (2150 BCE – 2026 CE)

© 2026 Yaroslav Marshalkin

Yaroslav Marshalkin

Corresponding author: marshalkin@gmail.com

Telegram: @MRSHLKN | GitHub: github.com/marshalkin-ux/paleoseismic-clustering

Abstract.

Whether large earthquakes cluster in space and time beyond chance is a foundational question in probabilistic seismic hazard assessment. We analyse a merged catalog of 4,418 $M \geq 6.5$ events (2150 BCE–2026 CE) from USGS ComCat, the ISC Bulletin, and NOAA NGDC ($M_c = 6.55$; $b = 0.911 \pm 0.018$). Using the Baiesi–Paczuski nearest-neighbor metric η with tectonic-path distance on the Bird (2003) plate-boundary graph, we identify 47 global seismic series across three temporal windows: 27 modern (1973–2026), 15 early instrumental (1900–1972), and 5 historical (pre-1900). Significance is established through: (i) a permutation test ($n=10,000$, $p < 0.0001$, $z = -6.17$); (ii) ETAS null-model validation (100 synthetic catalogs, $p_{\text{ETAS}} = 0.0000$, $\text{FPR} = 0/100$); and (iii) Benjamini–Hochberg FDR correction at $q = 0.05$ (45/47 series significant). The largest modern series, S170, comprises 46 events across 12 Flinn–Engdahl regions (2002–2023, $M_{\text{max}} = 9.1$), including the 2004 Sumatra–Andaman earthquake. Tectonic distance improves clustering sensitivity by $\sim 0.3 \log_{10} \eta$ units versus Euclidean distance.

Keywords: global seismicity; seismic series; earthquake clustering; tectonic distance; Baiesi–Paczuski metric; ETAS validation; false discovery rate; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that $M \geq 7$ clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global $M \geq 7$ and $M \geq 8$ rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. We test the hypothesis that multi-regional seismic series exist in a four-millennium catalog of $M \geq 6.5$ earthquakes, using an adapted η metric with tectonic distance along Bird (2003) plate boundaries. **Novelty.** To our knowledge, this is the first global application of nearest-neighbor clustering with tectonic-path distance across historical, early instrumental, and modern catalogs, combined with ETAS validation and FDR correction.

2. DATA AND METHODS

2.1 Catalog compilation

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records from ~2150 BCE). Duplicates were merged using ± 30 days and ≤ 50 km tolerance; source priority: ISC > USGS > NOAA. Only events with `quality_score` > 0.5 were retained.

Final catalog: 4,418 events $M \geq 6.5$ (2150 BCE – 2026 CE): 2,041 modern (1973–2026), 2,179 early instrumental (1900–1972), 67 historical (pre-1900).

2.1.1 Catalog completeness

Maximum-curvature analysis yields $M_c = 6.55$. Maximum-likelihood b -value from 1,688 events above M_c :

$$b = 0.911 \pm 0.018 \quad (n = 1,688 \text{ events})$$

2.2 Tectonic distance and connectivity metric

Tectonic distance r_{ij} is the shortest path between hypocenters along the Bird (2003) plate-boundary graph (20 key segments). Paths are computed with Dijkstra's algorithm (NetworkX).

For weakly connected pairs ($r_{\text{tect}} > 1.5 \times r_{\text{eucl}}$), a penalty $r_{ij} = 1.5 \times r_{\text{eucl}}$ is applied.

$$\eta_{ij} = t_{ij} \cdot r_{ij}^{1.6} \cdot 10^{(-b \cdot m_i)}$$

Following Baiesi & Paczuski [2004]; $d_f = 1.6$ (fractal dimension).

2.3 Series detection and validation

1. Declustering via Gardner–Knopoff [1974] to remove local aftershocks.
2. Nearest-neighbor forest: for each j , parent $i^* = \text{argmin } \eta_{ij}$ subject to $\eta_{ij} < \eta_0$.
3. Global series criterion: $N \geq 4$ events; $M \geq 6.5$; ≥ 3 Flinn–Engdahl regions.
4. Sliding windows (1, 2, 5 yr; 1-yr step); overlapping groups merged.

Permutation test: $n = 10,000$, $p < 0.0001$, $z = -6.17$ (modern). ETAS validation: 100 synthetic catalogs without long-range coupling, 0 false series, $p_{\text{ETAS}} = 0.0000$. FDR (Benjamini–Hochberg, $q = 0.05$): 45/47 series significant. Declustering: Gardner–Knopoff 98.8%; Zaliapin–Ben-Zion 100.0% independent.

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 global seismic series: 27 modern (1973–2026, $p < 0.0001$, $z = -6.17$), 15 early instrumental (1900–1972, $p = 0.007$, $z = -2.43$), and 5 historical ($p = 0.46$). The algorithm identifies 142 cluster candidates before window filtering and FDR correction.

ID	N	Regions	M_{max}	Period	Notes
S047	53	5	8.0	1982–2024	Western Pacific subduction corridor
S170	46	12	9.1	2002–2023	Sunda belt; Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	Western Pacific arc
S116	22	5	8.2	1993–2021	South Pacific multi-arc series
S191	15	4	8.4	2007–2022	Post-2004 activation episode

Table 1. Top five multi-regional seismic series (≥ 3 Flinn–Engdahl regions).

3.2 Spatial–temporal distribution

Elevated series activity occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic distance lowers the η_0 threshold by $\sim 0.3 \log_{10}$ units compared with Euclidean distance.

4. DISCUSSION AND CONCLUSIONS

Observed global series are incompatible with both a spatially inhomogeneous Poisson null hypothesis and ETAS catalogs that reproduce local aftershock structure without long-range coupling. FDR correction confirms that 45/47 identified series are not artifacts of multiple testing.

Tectonic-path distance is essential: Euclidean metrics underestimate connectivity along plate boundaries and overestimate links across rigid lithosphere.

1. A unified catalog of 4,418 $M \geq 6.5$ events contains 47 global seismic series; 27 modern series are significant at $p < 0.0001$ (Monte Carlo, $n = 10,000$).
2. ETAS validation (0/100 false positives) and FDR correction (45/47 at $q = 0.05$) confirm that series are not explained by randomness or local aftershock clustering alone.
3. Series S170 (46 events, 12 regions, 2002–2023, $M_{\max} = 9.1$) demonstrates long-term global activation along the Pacific Ring of Fire.
4. Tectonic distance increases η -metric sensitivity by $\sim 0.3 \log_{10}$ units relative to Euclidean distance.

Future work: Coulomb stress change (ΔCFS) analysis for S047, S170, and S095; focal mechanism constraints; ETAS extensions with explicit long-range kernels; Zenodo DOI release.

DATA AND CODE AVAILABILITY

Interactive presentation: marshalkin-ux.github.io/paleoseismic-clustering/ Source code: github.com/marshalkin-ux/paleoseismic-clustering. Results: [results/analysis_full_historical.json](#), [results/etas_validation.json](#), [results/fdr_correction_results.csv](#).

REFERENCES

1. Baiesi M., Paczuski M. (2004). Scale-free networks of earthquakes and aftershocks. *Phys. Rev. E*, 69, 066106.
2. Zaliapin I. et al. (2008). Clustering analysis of seismicity and aftershock identification. *Phys. Rev. Lett.*, 101, 018501.
3. Zaliapin I., Ben-Zion Y. (2013). Earthquake clusters in southern California I. *J. Geophys. Res.*, 118, 2847–2864.
4. Bird P. (2003). An updated digital model of plate boundaries. *Geochem. Geophys. Geosyst.*, 4(3), 1027.
5. Hill D.P. et al. (1993). Seismicity remotely triggered by the magnitude 7.3 Landers earthquake. *Science*, 260, 1617–1623.
6. Brodsky E.E., Prejean S.G. (2006). New constraints on mechanisms of remotely triggered seismicity. *J. Geophys. Res.*, 111, B06312.
7. Pollitz F.F. et al. (1998). Viscosity of oceanic asthenosphere inferred from remote triggering. *Science*, 280, 1245–1249.
8. Michael A.J. (2011). Random variability explains apparent global clustering of large earthquakes. *Geophys. Res. Lett.*, 38, L21301.
9. Shearer P.M., Stark P.B. (2012). Global risk of big earthquakes has not recently increased. *PNAS*, 109(3), 717–721.
10. Ogata Y. (1988). Statistical models for earthquake occurrences and residual analysis. *J. Amer. Stat. Assoc.*, 83, 9–27.
11. Benjamini Y., Hochberg Y. (1995). Controlling the false discovery rate. *J. Roy. Stat. Soc. B*, 57(1), 289–300.
12. Gardner J.K., Knopoff L. (1974). Is the sequence of earthquakes in Southern California Poissonian? *BSSA*, 64, 1363–1367.

13. Young J.B. et al. (1996). The Flinn–Engdahl regionalization scheme: the 1995 revision. *Phys. Earth Planet. Int.*, 96, 223–297.

Acknowledgments. The author acknowledges USGS, NOAA NGDC, and the ISC for maintaining open seismic catalogs. This work received no targeted funding.